Roll No. | | | | | | | |

Mid Semester Examination 2025

Name of the Program: M.Tech. Biomedical Engineering — Semester: I

Name of the Course: Nanotechnology in Medicine — Course Code: BEM 143

Time: 90 minutes — Maximum Marks: 50

Note:-

(iii) Answer all the questions by choosing any one of the sub-questions

(iv) Each question carries 10 marks

Q1	(10 marks)	CO2
(a)	Discuss change in physico-chemical properties of materials with their size reduction in nano dimensions.	
	OR	
(b)	Define fullerenes and discuss their applications in biomedical engineering.	
Q2	(10 marks)	CO2
(a)	What are different class of semiconducting nanoparticles? Briefly discuss their applications in biomedical engineering.	
	OR	
(b)	Define quantum dots. Briefly discuss the applications of carbon quantum dots in biomedical engineering.	
Q3	(10 marks)	CO2 & CO3
(a)	Describe the process of particle size determination using following techniques: (i) X-ray diffraction (ii) Zeta potential	
	OR	
(b)	With a suitable sketch, describe detailed process of fabricating crystalline thin film using chemical vapor deposition (CVD) technique.	
Q4	(10 marks)	CO3
(a)	What are carbon nanotubes (CNTs)? Discuss their types, basis of classification, and properties (mechanical, electrical, and thermal).	
	OR	
(b)	Briefly explain Arc Discharge and Laser Ablation methods of synthesizing CNTs.	
Q5	(10 marks)	CO3
(a)	Briefly describe Sol-Gel Technique of fabricating nanomaterials along with its general mechanism, reaction characteristics and advantages.	
	OR	
(b)	Discuss co-precipitation method of nanoparticle fabrication and Ostwald Ripening process.	